MODELING AND UNDERSTANDING MULTI-USER INTERACTION IN COMPLETE CORRIDOR INTERSECTIONS: A DIFFERENTIAL GAME OPTIMAL CONTROL APPROACH

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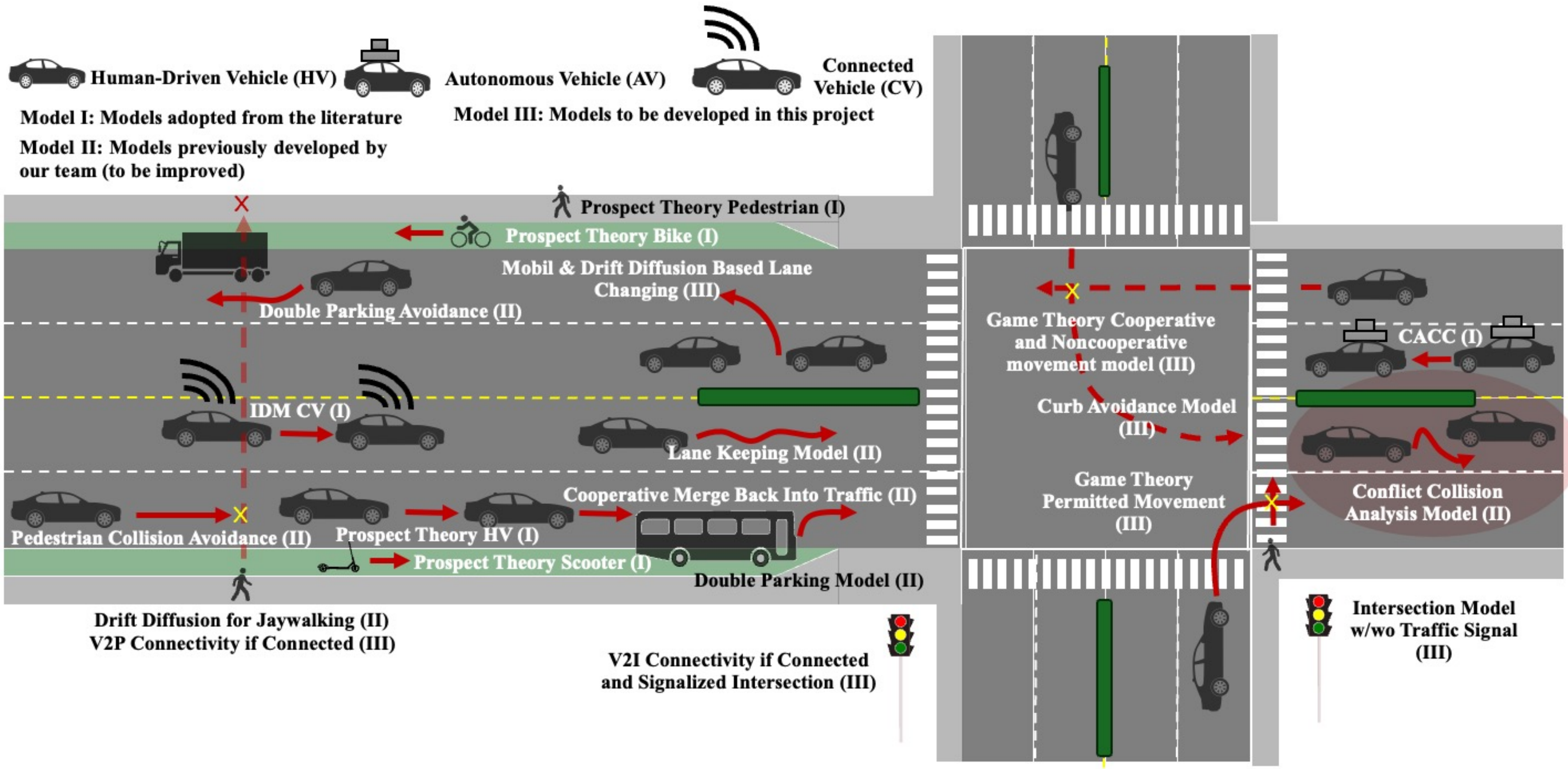
Corridor Model



Background

Hang et al. (2022) — Differential game for AV conflict resolution at unsignalized intersections

- **What they did:** Built a differential-game decision framework for AVs at unsignalized intersections, solving Nash/Stackelberg equilibria with safety/efficiency/aggressiveness terms and validating via hardware-in-the-loop.
- **Advantage:** Strong control/game-theoretic structure; real-time feasibility emphasis.
- **Limitations:** Primarily AV-focused; payoff parameters appear hand-designed (not calibrated from real trajectories).
- **Future work:** Calibrate aggressiveness/utility terms to real intersection data, extend to HV/AV mixed and richer actor sets.
- **Reference:** Hang, P., Huang, C., Hu, Z., & Lv, C. (2022). Driving conflict resolution of autonomous vehicles at unsignalized intersections: A differential game approach. IEEE/ASME Transactions on Mechatronics, 27(6), 5136–5146. https://doi.org/10.1109/TMECH.2022.3174273

Le Cleac'h et al. (2021) — LUCIDGames (inverse dynamic games + online estimation)

- **What they did:** Proposed online inverse dynamic games: estimate other agents' cost parameters from observed trajectories using a UKF, then plan in a receding-horizon game framework.
- **Advantage:** Directly addresses the "unknown objectives" problem; supports online adaptation.
- **Limitations:** Demonstrations are typically structured scenarios; applying to messy intersection data needs careful state/intent modeling.
- **Future work:** Robustness under partial observability, broader real-world datasets, multi-modal uncertainty.
- **Reference:** Le Cleac'h, S., Schwager, M., & Manchester, Z. (2021). LUCIDGames: Online unscented inverse dynamic games for adaptive trajectory prediction and planning. IEEE Robotics and Automation Letters, 6(3), 5485–5492. https://doi.org/10.1109/LRA.2021.3074880

Background

Peters et al. (2021) — Inverse objectives in continuous dynamic games from partial observations (RSS)

- **What they did:** Estimated parametric objectives in continuous dynamic games from noisy, partial state observations, coupling objective inference with Nash-equilibrium constraints for prediction.
- **Advantage:** Handles partial/noisy observation realistically; links inference to equilibrium consistency.
- **Limitations:** Demonstrations emphasize simulated traffic interactions; real intersections add occlusions, intent uncertainty, multimodality.
- **Future work**: Scaling to larger agent counts; richer behavioral classes; real-world intersection trajectory deployments.
- **APA:** Peters, L., Fridovich-Keil, D., Rubies-Royo, V., Tomlin, C. J., & Stachniss, C. (2021). Inferring objectives in continuous dynamic games from noise-corrupted partial state observations. In Robotics: Science and Systems XVII.

Peters et al. (2023) — Online & offline learning of objectives from partial observations (IJRR)

- **What they did:** Extended inverse objective learning to support both offline learning and online receding-horizon learning + prediction under partial observations.
- **Advantage:** Practical pipeline for continual adaptation; stronger bridge from learning → forecasting.
- **Limitations:** Still needs careful feature design / identifiability; real intersection data can violate model assumptions (unmodeled rules, pedestrians, signals).
- **Future work:** Better intent/state inference; integrating perception uncertainty; broader benchmarks.
- **Reference:** Peters, L., Rubies-Royo, V., Tomlin, C. J., Ferranti, L., Alonso-Mora, J., Stachniss, C., & Fridovich-Keil, D. (2023). Online and offline learning of player objectives from partial observations in dynamic games. International Journal of Robotics Research.

Background

Dai et al. (2022) — Calibrating driving preference/utility from observed trajectories (TRC)

- **What they did:** Formulated driver decision-making as utility maximization, then calibrated utility parameters from observed trajectories using a state-space/Kalman filter + MLE approach (shown on Sugiyama experiment trajectories).
- **Advantage:** Strong "inverse calibration" framing; yields interpretable preference parameters.
- **Limitations:** Demonstration setting is not intersection-specific; extending to intersections requires explicit conflict geometry and interaction structure.
- **Future work:** Intersection/mixed-traffic extensions; multi-agent calibration; richer datasets.
- **Reference:** Dai, Q., Shen, D., Wang, J., Huang, S., & Filev, D. (2022). Calibration of human driving behavior and preference using vehicle trajectory data. Transportation Research Part C: Emerging Technologies, 145, 103916. https://doi.org/10.1016/j.trc.2022.103916

Alsaleh & Sayed (2021) — Markov-game MA-AIRL for cyclist–pedestrian shared spaces (TRC)

- **What they did:** Modeled cyclist–pedestrian interactions as a Markov game, then used multi-agent adversarial IRL to recover reward functions and policies from real shared-space data.
- **Advantage:** Data-driven multi-agent interaction modeling; explicitly addresses non-stationary multi-agent behavior.
- **Limitations:** It's not a continuous-time differential game; transferability across sites can be challenging.
- **Future work:** Broader site transfer, richer agent heterogeneity, linking to AV planning.
- **Reference:** Alsaleh, R., & Sayed, T. (2021). Markov-game modeling of cyclist–pedestrian interactions in shared spaces: A multi-agent adversarial inverse reinforcement learning approach. Transportation Research Part C: Emerging Technologies.

Background

Lanzaro & Sayed (2025) — Vehicle–pedestrian interaction via Markov-game IRL across cities (Expert Systems with Applications)

- **What they did:** Used MA-AIRL to learn driver–pedestrian conflict interaction policies and tested transfer across multiple cities, showing transfer can raise risk if local behavior differs.
- **Advantage:** Direct focus on pedestrian interactions + cross-environment generalization testing.
- **Limitations:** Policies vary by environment; not a differential game trajectory-control formulation.
- **Future work:** Domain adaptation, local calibration, integration into AV safety systems.
- **Reference:** Lanzaro, G., & Sayed, T. (2025). Evaluating driver–pedestrian interaction behavior in different environments via Markov-game-based inverse reinforcement learning. Expert Systems with Applications, 260, 125405. https://doi.org/10.1016/j.eswa.2024.125405

Li et al. (2024) — Pedestrian trajectory prediction at right-turn unsignalized intersections using game theory (IEEE T-ITS)

- **What they did:** Proposed a pedestrian–vehicle game-theoretic model to predict pedestrian trajectories during right-turn conflicts at unsignalized intersections, using collected scene data.
- **Advantage:** Very intersection-specific and pedestrian-interaction centered.
- **Limitations:** It’s game-theoretic but not necessarily a continuous differential-game control model; extension to mixed autonomy may require AV/CV policy coupling.
- **Future work:** Mixed traffic (HV/AV), richer intent modeling, multi-site calibration.
- **Reference:** Li, W. (2024). A pedestrian trajectory prediction model for right-turn unsignalized intersections based on game theory. IEEE Transactions on Intelligent Transportation Systems.

Background

Johora et al. (2022) — Calibrated game-playing + social-force model for pedestrian–vehicle interactions

- **What they did:** Combined social-force interactions with game playing payoff matrices, and calibrated parametersusing real-world interaction scenarios (Hamburg Bergedorf station dataset).
- **Advantage:** Explicit calibration on real pedestrian–vehicle interactions; interpretable parameter categories (forces + safety + payoffs).
- **Limitations:** Not an intersection-trajectory differential game; applying to intersection maneuvers needs lane/priority/conflict-point structure.
- **Future work**: Intersection-specific geometry, mixed autonomy, scalable multi-agent calibration.
- **Reference:** Johora, F. T., et al. (2022). On intercultural transferability and calibration of game-theoretic social force models. arXiv preprint.

Chen et al. (2024) — Survey: game-theoretic human-like decision-making in mixed autonomy

- **What they did:** A broad survey of game-theoretic AV decision-making in mixed-autonomy traffic, summarizing challenges and future directions (personalization, uncertainty, validation).
- **Advantage:** Strong positioning map + explicitly identifies gaps (including limited real-data validation).
- **Limitations:** Not a new calibrated model; it's a review.
- **Future work:** Calls for more real-world datasets/validation, adaptive models, and social learning.
- **Reference:** : Chen, Q., Zhao, D., Yang, M., Shi, Y., & Liu, C. (2024). Autonomous vehicles in mixed-autonomy traffic: Game theoretic human-like decision making countermeasures. Complex Engineering Systems, 4, 25. https://doi.org/10.20517/ces.2024.69

Background

Li et al. (2025) — Human–vehicle interaction at right-turn unsignalized intersections (game theory + deep MaxEnt IRL)

- **What they did:** Proposed a data-driven framework for right-turn unsignalized intersections: pedestrian is primary agent, interaction described using game theory; uses IRL/MDP framing.

- **Advantage:** Right-turn unsignalized intersections + data-driven behavior.

- **Limitations:** Markov/IRL framing rather than differential-game control; integration into mixed HV/AV/CV still nontrivial.

- **Future work:** Mixed-traffic extension; coupling with AV planning/control layers.

- **Reference:** Li, W., et al. (2025). Simulation of human–vehicle interaction at right-turn unsignalized intersections: A game-theoretic deep maximum entropy inverse reinforcement learning method. Accident Analysis & Prevention.

Summary of Literature Review

#	Paper	Core Method	Data-Driven Calibration?	Pedestrians Included?	Mixed HV/AV?	Key Strength	Main Limitation	Gap
1	Hang et al. (2022)	Differential Game (Nash/Stackelberg)	No	No	AV-focused	Strong DG control structure; real-time feasibility	Hand-designed cost parameters	No HV, no pedestrians, no calibration
2	Le Cleac'h et al. (2021)	Inverse Dynamic Games (UKF)	Yes (online)	No	General agents	Learns unknown objectives online	Structured scenarios	Not intersection-specific; no pedestrians
3	Peters et al. (2021)	Inverse Objectives in DG	Yes	No	Generic agents	Handles noisy/partial observation	Mostly simulation-based	No real intersection deployment
4	Peters et al. (2023)	Online & Offline Objective Learning	Yes	No	Generic agents	Learning + prediction pipeline	Identifiability issues	Not intersection-focused; no multimodal conflict geometry
5	Dai et al. (2022)	Utility Calibration (State-Space + MLE)	Yes	No	HV only	Strong inverse calibration framing	Not intersection-specific	No DG; no multimodal interaction modeling
6	Alsaleh & Sayed (2021)	Markov Game + MA-AIRL	Yes	Cyclist–Pedestrian	No	Data-driven multi-agent IRL	Not continuous-time DG	No vehicle trajectory-level DG
7	Lanzaro & Sayed (2025)	Markov Game + IRL	Yes	Yes	No	Cross-city validation	Transferability issues	No DG trajectory control
8	Li et al. (2024)	Game-Theoretic Pedestrian Model	Partial	Yes	No	Intersection-specific pedestrian focus	Not continuous DG; limited AV coupling	No calibrated DG
9	Johora et al. (2022)	Social Force + Game Payoffs	Yes	Yes	No	Explicit calibration on real data	Not full DG trajectory model	No vehicle–vehicle DG interaction
10	Chen et al. (2024)	Survey (Mixed Autonomy)	No	Mixed	Conceptual	Identifies research gaps	Not empirical	Calls for real-data validation
11	Li et al. (2025)	Game Theory + Deep MaxEnt IRL	Yes	Yes	HV focus	Data-driven intersection modeling	MDP/IRL not DG control	No continuous-time DG + mixed autonomy

What's learned and what's missing?

What we learned?

- *Differential Game (DG) models*
- Strong theoretical foundation for strategic interaction
- Rarely calibrated using real intersection trajectory data
- Mostly AV-focused or small-scale interactions

- *IRL / Inverse Learning models*
- Data-driven and behaviorally grounded
- Typically rely on static or discrete-time decision layers
- Limited continuous-time trajectory-level equilibrium modeling
- Limited calibration

- *Social-force / Hybrid approaches*
- Capture pedestrian–vehicle interaction realism
- Do not solve multi-agent equilibrium or scalable mixed-traffic control

- *Across all models:*
- Limited modeling of mixed HV/AV/CV environments
- Limited corridor-level deployment
- Weak integration of safety-oriented conflict severity metrics
- Limited modeling of continuous negotiation and yielding dynamics
- Limited Calibration and Transferability

What's learned and what's missing?

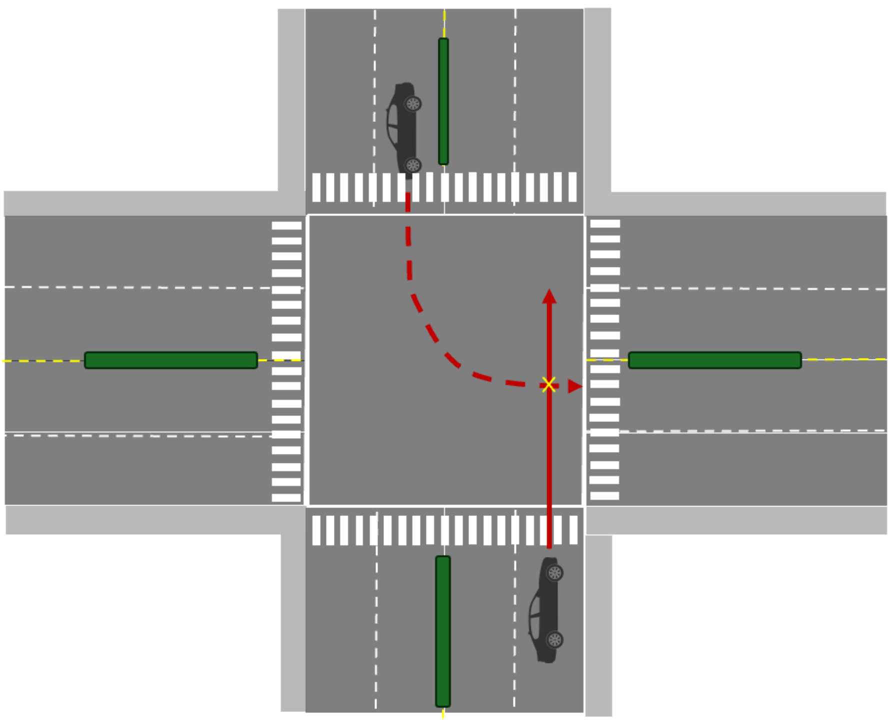
What missing?

- *What the Literature Lacks*
  - Mixed traffic representation across HV, AV, CV, pedestrians, and micromobility
  - Real-world calibration of strategic interaction parameters
  - Continuous-time modeling of trajectory-level
    - Negotiation
    - Yielding behavior
    - Gap acceptance
    - Conflict resolution
    - Car following
  - Uncertainty and noisy perception
  - Heterogeneity

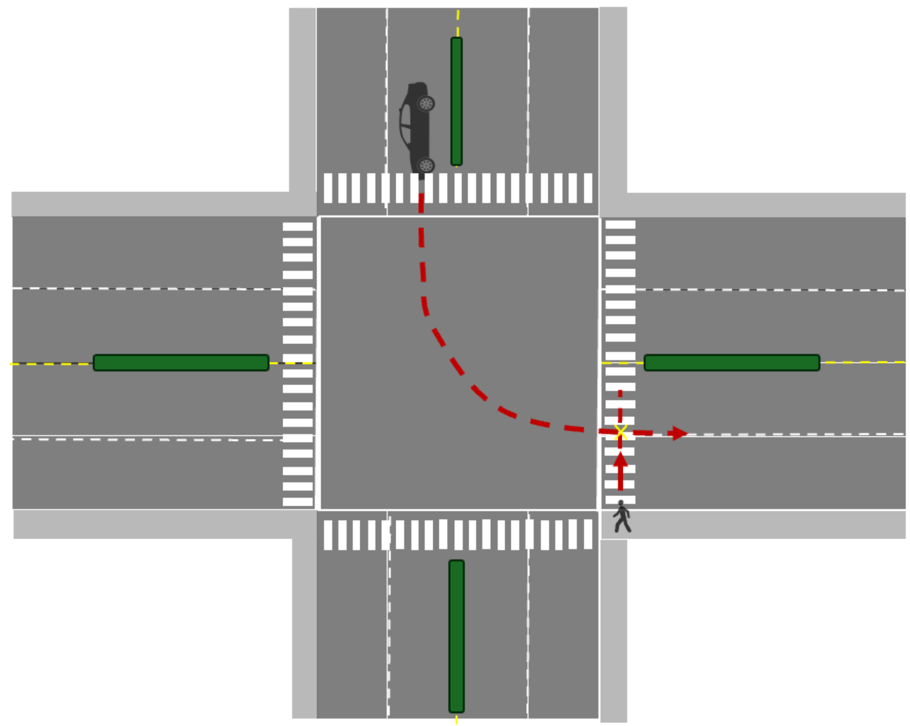
- *Our Direction*
  - Calibrate interaction behavior using real trajectory data
  - Model negotiation, yielding behavior, gap acceptance, and conflict resolution dynamics using continuous-time differential games
- *Beyond Isolated Intersection*
  - Shift beyond isolated intersection to corridor-level analysis
  - Evaluate how micro-interactions propagate along space and time
  - Integration of:
    - Lane allocation, curb management, signal strategies, geometric design
  - Use conflict-based safety metrics (TTC, Severity)
  - Unified framework for Complete Corridor Design

Problem Definitions

Permitted LT Conflicting with Thro Vehicle
(Gap acceptance, conflict resolution)

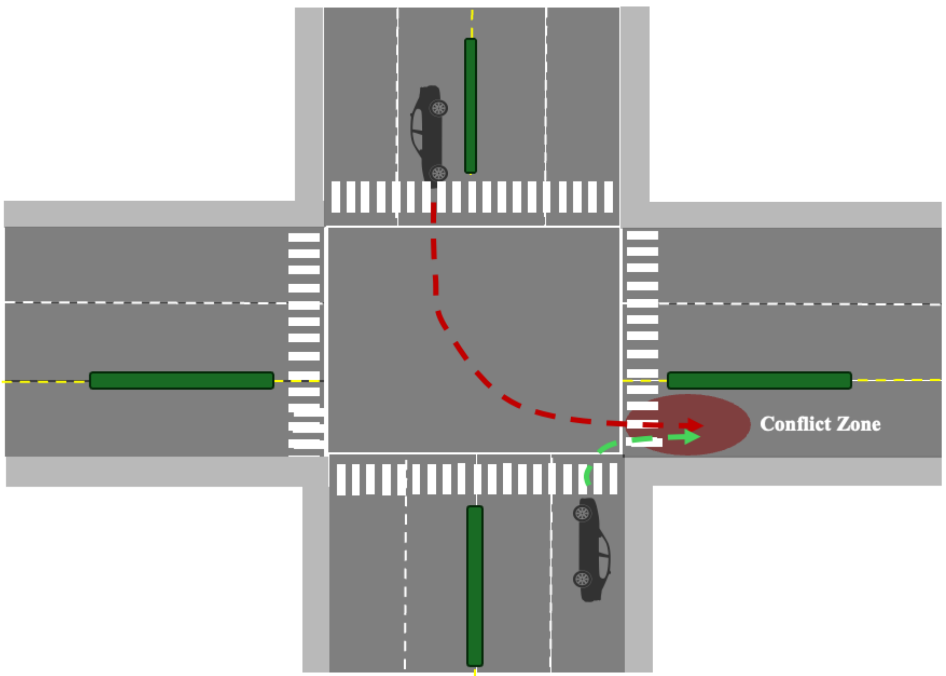


Permitted LT Conflicting with Ped, Bike, Scooters
(Gap acceptance, conflict resolution)

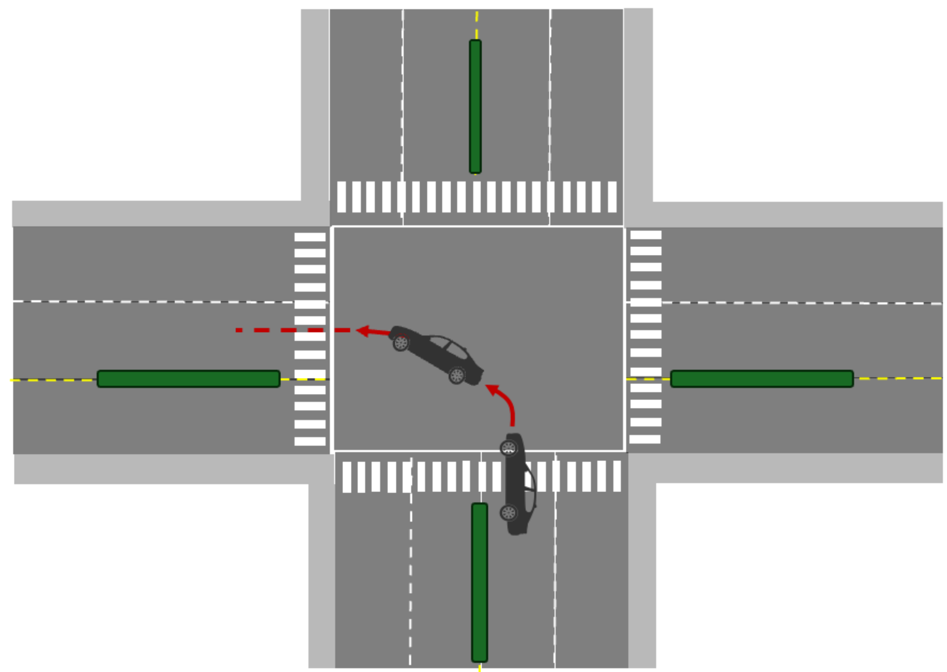


Problem Definitions

Merging into an Approach (Unsignalized Intersection)
(Negotiations, conflict resolution)



Car following Behavior within Intersection
(Car following)



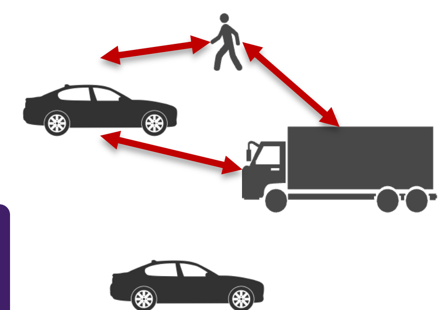
Problem Definitions - Framework

Parameter Calibration Layer

Game Theoretic Interaction Modelling

Optimal Control Formulation for each Individual Agent

β_i, ζ_i



Problem Definitions

Optimal Control Problem

- Vehicles trajectories are obtained by solving an optimal control problem.

State and Control Variables

- Vehicle i's position, steering angle, and speed are chosen as state variables.
  - $X_i(t) = \left[p_{xi}(t), p_{yi}(t), \theta_i(t), v_i(t)\right]^T$
- Vehicle i's acceleration and steering curvature are chosen as control variables.
  - $U_i(t) = [\kappa_i(t), a(t)]^T$ note: $\kappa_i(t) = \frac{1}{r_i(t)}$

System Dynamics

- $$\dot{X}_i = f\big(X_i(t), U_i(t)\big) = \begin{bmatrix} v_i(t)\cos\big(\theta_i(t)\big) \\ v_i(t)\, sin\big(\theta_i(t)\big) \\ v_i(t)\kappa_i(t) \\ a(t) \end{bmatrix}$$

Cost Function Definition

- $J_i(U_i, X_{-i}) = \psi\big(X_i(T)\big) - \int_0^T L(X_i(t), X_{-i}(t-\tau), U_i(t), t)\, dt$
  - $L(X_i(t), X_{-i}(t), U_i(t), t)$: Running Cost
  - $\psi\big(X_i(T)\big)$: Terminal Cost
- $\psi\big(X_i(T)\big) = \frac{1}{2}\mathrm{b}\left|\left|X_i(T) - X_{id}\right|\right|^2$
  - In which the $X_i(T)$ is the terminal location of the vehicle at intersection, and X_{id} is the desired final location.
- $L(X_i(t), X_{-i}(t-\tau), U_i(t), t) = \int_0^T \Big(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t)$
  $+ \frac{\beta_3}{2} a_i^2(t) + \frac{\beta_4}{2} v_i^2(t) e^{-\zeta_1\, \phi_i(t)\, TTC_i(t)}$
  $+ \Sigma_j \frac{\beta_5}{2} m_i m_j v_{relij}^2(t) e^{-\zeta_2 TTC_{ij}(t)}\Big)\mathrm{dt}$
- β_1: Parameter that shows sensitivity of the vehicle to the travel time
- β_2: Parameter that shows sensitivity of the vehicle to the latitudinal or centripetal acceleration (Smoothness and Comfort)
  - $\frac{\beta_2}{2}\kappa_i(t)^2 v_i(t)^4 = \frac{\beta_2}{2} a_{ci}^2(t) = \frac{\beta_2}{2}\left(\frac{v_i(t)^2}{r_i(t)}\right)^2$
    $= \frac{\beta_2}{2}(\kappa_i(t) v_i(t)^2)^2$
- β_3: Parameter that shows sensitivity of the vehicle to the longitudinal acceleration (Smoothness and Comfort)

1) Zhao, J., Knoop, V. L., & Wang, M. (2020). Two-dimensional vehicular movement modelling at intersections based on optimal control. Transportation Research Part B: Methodological, 138, 1-22.
2) Zhao, J., Knoop, V. L., & Wang, M. (2023). Microscopic traffic modeling inside intersections: Interactions between drivers. Transportation science, 57(1), 135-155.

Problem Definitions

- $\int_0^T L(X_i(t), X_{-i}, (t-\tau), U_i(t), t) = \int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t) + \frac{\beta_3}{2} a_i^2(t) + \frac{\beta_4}{2} v_i^2(t) e^{-\zeta_1 \phi_i(t)\, TTC_i(t)} + \sum_j \frac{\beta_5}{2} m_i m_j v_{relij}^2(t) e^{-\zeta_2 TTC_{ij}(t)}\right)\mathrm{dt}$

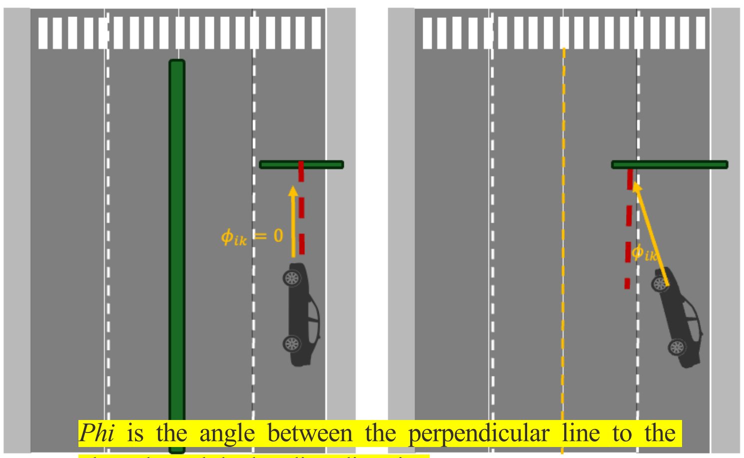
Sensitivity to the travel time

Sensitivity to the latitudinal acceleration

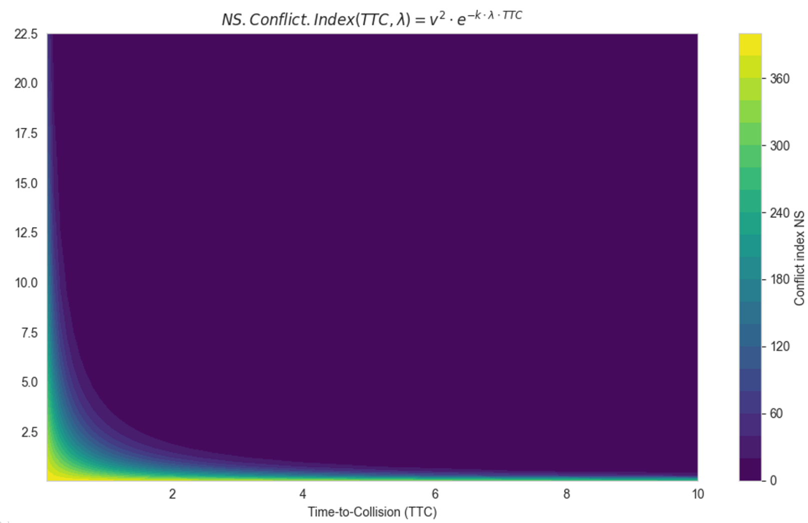
Sensitivity to the longitudinal acceleration

Conflict with the curbs, refuge island, (Stationary Obstacles)

Conflict with the moving obstacles (Non-stationary Obstacles)



Phi is the angle between the perpendicular line to the obstacle and the heading direction



Problem Definitions

Generalized Total Time to Collision (GTTC) Mathematical Formulation

- $d_{ij}(t) = \sqrt{\left(c_i(t) - c_j(t)\right)^T \left(c_i(t) - c_j(t)\right)}$
    - $c_i(t)$: Agent i's position at time t
    - $d_{ij}(t)$: Distance between agent i and j
- $\frac{d}{dt}\left(d_{ij}^2\right) = \frac{d}{dt}\left(\left(c_i - c_j\right)^T \left(c_i - c_j\right)\right)$
- $\dot{d_{ij}} = \frac{\left(c_i - c_j\right)^T \left(v_i - v_j\right)}{d_{ij}}$
- $GTTC = -\frac{d_{ij}}{\dot{d_{ij}}}$
- $d_{ij}^{corrected} = \sqrt{\left(o_i - o_j\right)^T \left(o_i - o_j\right)} - 0.5\left(L_1 + L_2\right)$
- $TTC = -\frac{d_{ij}}{\dot{d_{ij}}} = \frac{\sqrt{\left(o_i - o_j\right)^T \left(o_i - o_j\right)} - 0.5\left(L_1 + L_2\right)}{\frac{\left(o_i - o_j\right)^T \left(v_i - v_j\right)}{\sqrt{\left(o_i - o_j\right)^T \left(o_i - o_j\right)}}}$

Generalized Time-to-Collision (GTTC) algorithm embedded in the simulator

i. Relative position calculation
   - *i.* $\boldsymbol{r = o_i - o_j}$

ii. Relative speed calculation
   - *i.* $\boldsymbol{v_{rel} = v_i - v_j}$

iii. Separation distance computation
   - *i.* $\boldsymbol{d_{ij} = \left|\left|r\right|\right| = \sqrt{r^T r}}$

iv. Compute the approach rate
   - *i.* $\boldsymbol{\dot{d_{ij}} = \frac{r^T v_{rel}}{\left|\left|r\right|\right|}}$

v. Adjust for the agents' length
   - *i.* $\boldsymbol{d_{ij,adjusted} = d_{ij} - 0.5(L_1 + L_2)}$

vi. Generalized TTC calculation
   - *i.* $\boldsymbol{GTTC = -\frac{d_{ij,adjusted}}{\dot{d_{ij}}}}$

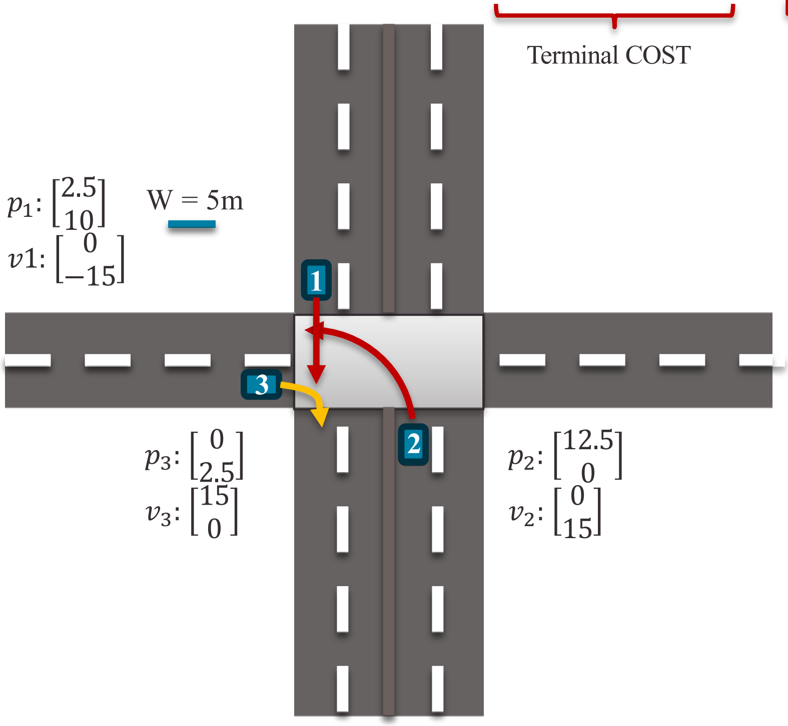
Note 1: Negative values indicate that the agents are moving away from each other.

Note 2: A conflict occurs when the GTTC falls below a specific threshold.

Problem Definitions

Cost Function

- $$J_i(X_i(t), X_{-i}(t-\tau), U_i(t), t) = \underbrace{-\frac{1}{2}\mathrm{b}\left|\left|X_i(T) - X_{id}\right|\right|^2}_{\text{Terminal COST}} - \underbrace{\int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t) + \frac{\beta_3}{2}a_i^2(t) + \frac{\beta_4}{2}v_i^2(t)e^{-\zeta_1\,\phi_i(t)\,TTC_i(t)} + \sum_j \frac{\beta_5}{2}m_i m_j v_{relij}^2(t)e^{-\zeta_2 TTC_i(t)}\right)\mathrm{dt}}_{\text{Running Cost}}$$



Subject to:

- $a_{min} \leq a(t) \leq a_{max}$
- $\kappa_{min} \leq \kappa(t) \leq \kappa_{max}$
- $v_{min} \leq v(t) \leq v_{max}$

Single vehicle case solution

- *Fairly simple to solve using Pontryagin's Minimum Principle*
- $\mathcal{H}(\boldsymbol{X}, \boldsymbol{U}, \boldsymbol{\lambda}) = L(\boldsymbol{X}, \boldsymbol{U}) + \boldsymbol{\lambda}^T f(\boldsymbol{X}, \boldsymbol{U})$
- *Costate dynamics:* $\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial \mathcal{H}^T}{\partial \boldsymbol{X}}$
- *Optimal control law:* $\boldsymbol{u}^*(t) = argmin_{u \in U}\, \mathcal{H}\left(\boldsymbol{X}(t), \boldsymbol{U}(t), \boldsymbol{\lambda}(t)\right)$
- *Final state is free thus:* $\boldsymbol{\lambda}(T) = \nabla\Phi\left(\boldsymbol{X}_i(T)\right)$

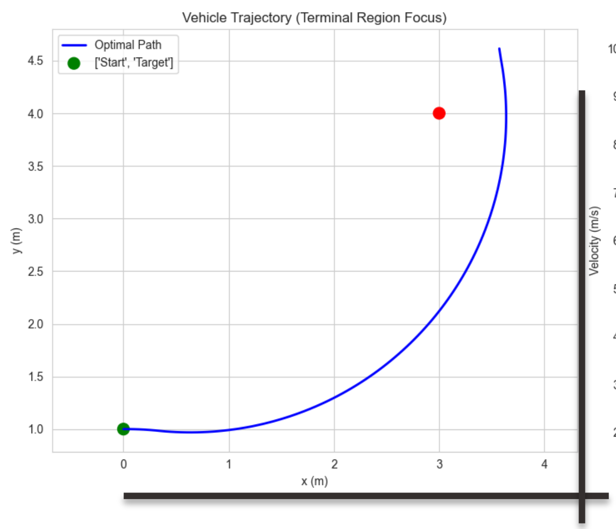
Problem Definitions

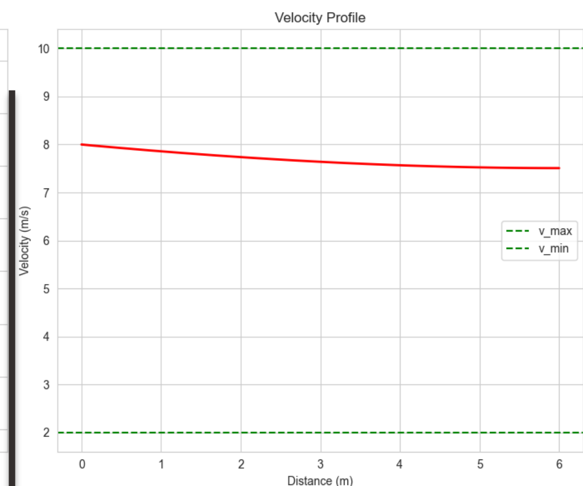
Cost Function

- $$J_i(X_i(t), X_{-i}(t-\tau), U_i(t), t) = \frac{1}{2}\mathrm{b}\left\|X_i(T) - X_d\right\|^2 + \int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa(t)^2 v(t)^4 + \frac{\beta_3}{2}a(t)^2 + \frac{\beta_4}{2}v(t)^2 e^{-\zeta_1 \phi_i(t) TTC(t)} + \sum_j \frac{\beta_5}{2} m_i m_j v_{rel}^2(t) e^{-\zeta_2 TTC(t)}\right) \mathrm{dt}$$

Subject to:

- $a_{min} \leq a(t) \leq a_{max}$
- $\kappa_{min} \leq \kappa(t) \leq \kappa_{max}$
- $v_{min} \leq v(t) \leq v_{max}$





Problem Definitions

Optimal Control Problem

- Vehicles' trajectories are obtained by solving an optimal control problem.

Interaction Between Vehicles is modelled with Non-Cooperative Differential Game

- For HV-HV modelling, the Open Loop Nash Equilibrium solution is used to obtain the trajectories, though the rolling horizon approach is used to implement only the first of the planned trajectories, and the process is repeated. Similarly, different game settings can be set based on different agent types.

State and Control Variables

- Vehicle i's position, steering angle, and speed are chosen as state variables.
  - $X_i(t) = \left[p_{xi}(t), p_{yi}(t), \theta_i(t), v_i(t)\right]^T$
- Vehicle i's acceleration and steering curvature are chosen as control variables.
  - $U_i(t) = [\kappa_i(t), a(t)]^T$ note: $\kappa_i(t) = \frac{1}{r_i(t)}$

Cost Function

- $J_i(X_i(t), X_{-i}(t-\tau), U_i(t), t) = -\frac{1}{2}\mathrm{b}\left|\left|X_i(T) - X_{id}\right|\right|^2 - \int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t) + \frac{\beta_3}{2}a_i^2(t) + \frac{\beta_4}{2}v_i^2(t)e^{-\zeta_1\,\phi_{ik}(t)\,TTC_{ik}(t)} + \sum_j \frac{\beta_5}{2}m_i m_j v_{relij}^2(t)e^{-\zeta_2 TTC_{ij}(t)}\right)\mathrm{dt}$

System Dynamics

- $$\dot{X}_i = f\left(X_i(t), U_i(t)\right) = \begin{bmatrix} v_i(t)\cos\left(\theta_i(t)\right) \\ v_i(t)\,sin\left(\theta_i(t)\right) \\ v_i(t)\kappa_i(t) \\ a(t) \end{bmatrix}$$

Subject to:

- $a_{min} \leq a(t) \leq a_{max}$
- $\kappa_{min} \leq \kappa(t) \leq \kappa_{max}$
- $v_{min} \leq v(t) \leq v_{max}$

Problem Definitions

Cost Function

- $J_i(X_i(t), X_{-i}(t), U_i(t), t) = -\frac{1}{2}\mathrm{b}\left\|X_i(T) - X_{id}\right\|^2 - \int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t) + \frac{\beta_3}{2}a_i^2(t) + \frac{\beta_4}{2}v_i^2(t)e^{-\zeta_1\,\phi_{ik}(t)\,TTC_{ik}(t)} + \sum_j \frac{\beta_5}{2}m_i m_j v_{relij}^2(t)e^{-\zeta_2 TTC_{ij}(t)}\right)\mathrm{dt}$

State and Control Variables

- $X_i(t) = \left[p_{xi}(t), p_{yi}(t), \theta_i(t), v_i(t)\right]^T$
- $U_i(t) = [\kappa_i(t), a(t)]^T$ note: $\kappa_i(t) = \frac{1}{r_i(t)}$

System Dynamics

- $$\dot{X}_i = f\big(X_i(t), U_i(t)\big) = \begin{bmatrix} v_i(t)\cos\big(\theta_i(t)\big) \\ v_i(t)\,sin\big(\theta_i(t)\big) \\ v_i(t)\kappa_i(t) \\ a_i(t) \end{bmatrix}$$

Subject to:

- $a_{min} \le a_i(t) \le a_{max}$
- $\kappa_{min} \le \kappa_i(t) \le \kappa_{max}$
- $v_{min} \le v_i(t) \le v_{max}$
- $X_i(t=0) = X_{i0}$

Open-Loop Nash Equilibrium

- A set of control functions $t \to \left(u_1^*(t), \dots, u_{i-1}^*(t), u_i^*(t), u_{i+1}^*(t), \dots, u_n^*(t)\right)$ is a Nash Equilibrium for the previously mentioned game within the class of open-loop strategies if the following holds.
- **i)** The control $u_1^*(.)$ provides a solution to the optimal control problem for player 1:
  - Maximize: $J_1(u_1, \dots, u_{i-1}^*, u_i^*, u_{i+1}^*, \dots, u_n^*)$
  - Over all controls $u_1(.)$ for the system with previously mentioned dynamics.
- **ii)** The control $u_n^*(.)$ provides a solution to the optimal control problem for player n:
  - Maximize: $J_1(u_1^*, \dots, u_{i-1}^*, u_i^*, u_{i+1}^*, \dots, u_n)$
  - Over all controls $u_n(.)$ for the system with previously mentioned dynamics.
- **iii)** The control $u_i^*(.)$ provides a solution to the optimal control problem for player i:
  - Maximize: $J_1(u_1^*, \dots, u_{i-1}^*, u_i, u_{i+1}^*, \dots, u_n^*)$
  - Over all controls $u_i(.)$ for the system with previously mentioned dynamics.
- All functions $f, \psi_1, \dots, \psi_{i-1}, \psi_i, \psi_{i+1}, \dots, \psi_n, L_1, \dots, L_{i-1}, L_i, L_{i+1}, \dots, L_n$ are continuously differentiable, thus the necessary conditions for optimality can be provided by Pontryagin Maximum Principle.

Problem Definitions

Cost Function

- $J_i(X_i(t), X_{-i}(t-\tau), U_i(t), t) = -\frac{1}{2}\mathrm{b}\left|\left|X_i(T) - X_{id}\right|\right|^2 - \int_0^T \left(\beta_1 + \frac{\beta_2}{2}\kappa_i(t)^2 v_i^4(t) + \frac{\beta_3}{2}a_i^2(t) + \frac{\beta_4}{2}v_i^2(t)e^{-\zeta_1\,\phi_{ik}(t)\,TTC_i(t)} + \sum_j \frac{\beta_5}{2}m_i m_j v_{relij}^2(t)e^{-\zeta_2 TTC_{ij}(t)}\right)\mathrm{dt}$

State and Control Variables

- $X_i(t) = \left[p_{xi}(t), p_{yi}(t), \theta_i(t), v_i(t)\right]^T$
- $U_i(t) = [\kappa_i(t), a(t)]^T$ note: $\kappa_i(t) = \frac{1}{r_i(t)}$

System Dynamics

- $\dot{X}_i = f\left(X_i(t), U_i(t)\right) = \begin{bmatrix} v_i(t)\cos\left(\theta_i(t)\right) \\ v_i(t)\,sin\left(\theta_i(t)\right) \\ v_i(t)\kappa_i(t) \\ a_i(t) \end{bmatrix}$

Subject to:

- $a_{min} \le a_i(t) \le a_{max}$
- $\kappa_{min} \le \kappa_i(t) \le \kappa_{max}$
- $v_{min} \le v_i(t) \le v_{max}$
- $X_i(t=0) = X_{i0}$

Open-Loop Nash Equilibrium

- Applying PMP will lead to the following set of necessary conditions for optimality:

- **Hamiltonian:**
  - $\mathcal{H}_i(t, X_i, X_{-i}^{-\tau}, U_i, \lambda_i) = \lambda_i^T f(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*) - L(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*)$
  - $U_i^* = argmax_{U_i}\begin{pmatrix}\lambda_i^T f(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*) - \\ L(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*)\end{pmatrix}$
  - $\forall i \in \mathbb{N}$
- **System of Equations:**
  - $\dot{X}_i(t) = f(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*)$
  - $\dot{\lambda}_i(t) = -\lambda_i\frac{\partial f}{\partial X_i}(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*)$

  $+\ \frac{\partial L}{\partial X_i}(t, X_i, X_{-i}^{-\tau}, U_1^*, \ldots, U_{i-1}^*, U_i, U_{i+1}^*, \ldots, U_n^*)$
- **Terminal Conditions:**
  - $X_i(t=0) = X_{i0}$
  - $a_{min} \le a_i(t) \le a_{max}$
  - $\kappa_{min} \le \kappa_i(t) \le \kappa_{max}$
  - $v_{min} \le v_i(t) \le v_{max}$
  - $\lambda_i(T) = \nabla\psi_i\left(X_i(T)\right) = -\frac{\partial}{\partial X_i(T)}\frac{1}{2}b(X_i(T) - X_{id})^T(X_i(T) - X_{id})$

  $= b(X_i(T) - X_{id})$

Solution Method

Nonlinear general-sum differential game

- We have N interacting agents with nonlinear dynamics and agent-specific costs:
- Dynamics:
  - $\dot{x} = f(t, x, u_{1:N})$
  - $J_i(u_{1:N}(.)) = \int_0^T g_i(t, x(t), u_{1:N}(t))\, dt$
- We are looking for **time-varying state-feedback strategies** $\gamma_i(t, x)$

Iterative Linear Quadratic Games (ILQR)

- LQ game:
  - Linear Dynamics
  - Quadratic Cost
- Assume N players:
- $\dot{x} = Ax + \sum_{i=1}^{N} B_i u_i$
- $J_i = x_{iT}^T Q_{iT} x_{iT} + \int_0^T x_{it}^T Q_{it} x_{it} + \sum_{j=1}^{N} u_{it}^T R_{ijt} u_{it}$
- LQ game can be solved as Feedback or Open-loop Nash Equilibrium

Fridovich-Keil, D., Driggs-Campbell, K., Gillula, J. H., & Tomlin, C. J. (2020).
Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games. *IEEE International Conference on Robotics and Automation*
Başar, T., & Olsder, G. J. (1999).
Dynamic noncooperative game theory (2nd ed.). SIAM.
Engwerda, J. C. (2005).
LQ dynamic optimization and differential games. Wiley.

Solution Method

Iterative Linear Quadratic Games (ILQR)

- By repeatedly solving local LQ game approximations around the current trajectory:
- At iteration k:
  - Rollout current strategies: trajectory iterate
    - $\xi^k = \{\hat{x}(t), \hat{u}_{1:N}(t)\}_{t\in[0,T]}$
  - Linearize dynamic around ξ^k:
    - $\dot{\delta x}(t) \approx A(t)\delta x(t) + \sum_{i=1}^{N} B_i(t)\delta u_i(t)$
- Quadraticize each player's running cost around ξ^k:
  - $g_i(.) \approx g_i(\hat{.}) + \frac{1}{2}\delta x^T(Q_i\delta x + 2l_i) + \frac{1}{2}\sum_{j=1}^{N}\delta u^T(R_{ij}\delta u_j + 2r_{ij})$
- Solve the resulting finite-horizon LQ game: new candidate feedback strategies (affine):
  - $\widetilde{\gamma_i^k}(t,x) = \widehat{u_i}(t) - P_i^k(t)\delta x(t) - \eta\alpha_i^k(t)$
- $\eta \in (0,1]$
  - $\eta = 0$: Stay at previous open-loop rollout
  - $\eta = 1$: Take full LQ-game update
    - This damping is used cause new trajectory may go out of the local region and LQ is valid

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Appendix A

Open-loop Nash (LQ)

- Each agent commits to a control plan that depends on time (and maybe initial state), not the evolving state:
  - $u_t^{i*} = u_t^{i*}(t, x_0)$
- **Pontryagin minimum principle** with a Hamiltonian H_t^i is used.
- Define: $\Lambda_t = I + \sum_{j=1}^{N} B_{j,t} R_{jj,t}^{-1} B_{j,t}^T M_{t+1}^j$
- And backward recursions:
  - $M_t^i = Q_t^i + A_t^T M_{t+1}^i \Lambda_t^{-1} A_t$
  - $m_t^i = A_t^T \left(m_{t+1}^i - M_{t+1}^i \Lambda_t^{-1} \sum_j B_{j,t} R_{jj,t}^{-1} \left(B_{j,t}^T m_{t+1}^j + r_{jj,t}\right)\right) + q_t^i$
- Then the Nash open-loop control and forward state update:
  - $u_t^{i*} = -R_{ii,t}^{-1}\left(B_{i,t}^T\left(M_{t+1}^i x_{t+1}^* + m_{t+1}^i\right) + r_{ii,t}\right)$
  - $x_{t+1}^* = \Lambda_t^{-1}\left(A_t x_t^* - \sum_j B_{j,t} R_{jj}^{-1}\left(B_{j,t}^T m_{t+1}^j + r_{ii,j}\right)\right)$

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Appendix B

Feedback Nash (LQ)

- Each agent uses the current state x_t :
  - $u_t^{i*} = \gamma_t^i(x_t)$
- Assume quadratic value:
  - $V_t^i(x_t) = \frac{1}{2} x_t^T Z_t^i x_t + (\zeta_t^i)^T x_t + n_t^i$
- Optimal control affine:
  - $u_t^{i*} = -P_t^i x_t - \alpha_t^i$
- The gains P_t^i and offsets α_t^i are found by solving coupled linear systems (one for P, one for α):
  - $(R_{ii,t} + B_{i,t}^T Z_{t+1}^i B_{i,t}) P_t^i + B_{i,t}^T Z_{t+1}^i \sum_{j \neq i} B_{j,t} P_t^j = B_{i,t}^T Z_{t+1}^i A_t$
  - $(R_{ii,t} + B_{i,t}^T Z_{t+1}^i B_{i,t}) \alpha_t^i + B_{i,t}^T Z_{t+1}^i \sum_{j \neq i} B_{j,t} \alpha_t^j = B_{i,t}^T \zeta_{t+1}^i + r_{ii,t}$
- Then "Riccati-like" backward recursions update the value parameters:
  - $Z_t^i = Q_t^i + \sum_j P_t^{jT} R_{ij,t} P_t^j + F_t^T Z_{t+1}^i F_t$
  - $\zeta_t^i = q_t^i + \sum_j (P_t^{jT} R_{ij,t} \alpha_t^j - P_t^{jT} r_{ij,t}) + F_t^T (\zeta_{t+1}^i + Z_{t+1}^i \beta_t)$
- With:
  - $F_t = A_t - \sum_j B_{j,t} P_t^j$
  - $\beta_t = -\sum_j B_{j,t} \alpha_t^j$

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Overview of Continuous Time LQR

Lists of Contents

- Continuous-time LQR formulation
- Dynamic programming solution
- Hamiltonian system & boundary value problem
- Infinite horizon LQR
- Algebraic Riccati Equation (ARE)

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Overview of Continuous Time LQR

What question are we trying to answer with LQR?

- How do we control the system optimally while:
  - The state stays small (safe, stable, efficient)
  - The control effort is not too large (Realistic, smooth)

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Overview of Continuous Time LQR

System Dynamics:

- $\dot{x}(t) = Ax(t) + Bu(t), x(0) = x_0$
    - $x(t) \in \mathbb{R}^n \rightarrow state\ vector$
    - $u(t) \in \mathbb{R}^m \rightarrow control\ input$
    - $A \in \mathbb{R}^{n \times n} \rightarrow system\ matrix$
    - $B \in \mathbb{R}^{n \times m} \rightarrow input\ matrix$
    - $\dot{x}(t) \in \mathbb{R}^n$
    - For the dimension look that they should be summable.

Control Objective

We choose a function:

- $u: [0, T] \rightarrow \mathbb{R}^m$
- This means that at every time t, we pick a control vector $u(t)$

Cost Function

$$J = \int_0^T (x(\tau)^T Q x(\tau) + u(\tau) R u(\tau)^T d\tau) + x(T)^T Q_f x(T)$$

- $Q \succ= 0 \in \mathbb{R}^{n \times n} \rightarrow symmetric\ state\ cost$
- $Q_f \succ= 0 \in \mathbb{R}^{n \times n} \rightarrow symmetric\ final\ state\ cost$
- $R \succ 0 \in \mathbb{R}^{m \times m} \rightarrow symmetric\ control\ cost$

Overview of Continuous Time LQR

Mini Numerical Example

- 2D state, 1D control
- Let:
  - $x(t) = \begin{bmatrix} position \\ velocity \end{bmatrix} \in \mathbb{R}^2$
  - $u(t) = acceleration \in \mathbb{R}$
- **System Matrices:**
  - $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{2\times2}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{R}^{2\times1}$
- **Cost matrices**
  - $Q = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2\times2}, R = [1]$

Overview of Continuous Time LQR

Let's use some DP for the solution, shall we?

- Instead of solving from $0 \to T$, define: $V_t(z)$
- Definition:
  - $V_t(z) = \min_u \int_t^T \left(x(\tau)^T Q x(\tau) + u(\tau)^T R u(\tau)\right) d\tau + x(T)^T Q_f x(T)$
- Subject to:
  - $x(t) = z, \dot{x}(t) = Ax(t) + Bu(t)$
- $z \in \mathbb{R}^n$
- $V_t(z) \in \mathbb{R}$
- $x(\tau) \in \mathbb{R}^n$
- $u(\tau) \in \mathbb{R}^m$
- $Q, Q_f \in \mathbb{R}^{n \times n}, R \in \mathbb{R}^{m \times m}$

- $V_T(z) = z^T Q_f z$

Overview of Continuous Time LQR

Let's use some DP for the solution, shall we?

- Now, let's look at a very small interval $[t, t+h]$
- We need approximations, and use the dynamic programming idea.
- At time t, suppose:
  - $x(t) = z,\ z \in \mathbb{R}^n$
  - $u(t) = w \in \mathbb{R}^m$
    - We assume this is constant over $[t, t+h]$
- Accordingly, the cost in this interval is:
  - $\int_t^{t+h} \left(x(\tau)^T Q x(\tau) + u(\tau)^T R u(\tau)\right) d\tau = \int_t^{t+h} (z^T Q z + w^T R w d\tau) = h(z^T Q z + w^T R w)$
  - $\dot{x}(t) = Ax(t) + Bu(t) \rightarrow x(t+h) = x(t) + \dot{x}(t)\delta t = z + h(Az + Bw)$
- So, the new state summary: $x(t+h) = z + h(Az + Bw), J(t+h) = h(z^T Q z + w^T R w)$

Overview of Continuous Time LQR

Let's use some DP for the solution, shall we?

- So, the new state summary: $x(t+h) = z + h(Az + Bw), J(t+h) = h(z^T Qz + w^T Rw)$
- Now ***What's the cost to go from t+h***: Since the value function if quadratic in state:
  - $V_t(z) = z^T P_t z \;\rightarrow V_t(z)\ is\ quadratic$ – Can be proven via backward recursion
- $V_{t+h}(z) = \; z^T P_{t+h} z \;\rightarrow$ plug in new state $\rightarrow V(t+h)\big(z + h(Az + Bw)\big) \rightarrow P_{t+h} = P_t + h\dot{P} \rightarrow h \rightarrow V(t+h)\big(z + h(Az + Bw)\big) = \big(z + h(Az + Bw)\big)^T \big(P_t + h\dot{P}\big)\big(z + h(Az + Bw)\big) =$
  - $z^T P_t z + h\big[(Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\big]$
- $Total\ Cost = instantanous\ cost + cost\ to\ go = h(z^T Qz + w^T Rw) + z^T P_t z + h\big[(Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\big]$
- $Total\ Cost = z^T P_t z + h\big[z^T Qz + w^T Rw + (Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\big]$
  - *We didn't consider higher order terms with h*

Overview of Continuous Time LQR

Let's use some DP for the solution, shall we?

- Minimize the V with respect to the control variable w:
  - $\frac{d}{dw}\left(z^T P_t z + h\left[z^T Q z + w^T R w + (Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\right]\right) =$
  - $\frac{d}{dw}\left(w^T R w + (Az + Bw)^T P_t z + z^T P_t (Az + Bw)\right) \rightarrow z^T P_t (Az + Bw)$ is a scalar and a scaler is equal to its transpose
  - $\frac{d}{dw}\left(w^T R w + 2z^T P_t (Az + Bw)\right) = 2Rw + 2B^T P_t^T z = 0 \rightarrow w = -R^{-1} B^T P_t^T z \rightarrow P_t = P_t^T \rightarrow w^* = -R^{-1} B^T P_t z$
  - $R^{-1} \in \mathbb{R}^{m \times m}, B^T \in \mathbb{R}^{m \times n}, P_t^T z \in \mathbb{R}^n \rightarrow w^* \in \mathbb{R}^m$

Overview of Continuous Time LQR

Let's use some DP for the solution, shall we?

- $V_t(z) = immidiate\ cost + V_{t+h}(z) = z^T P_t z = z^T P_t z + h\left[z^T Q z + w^T R w + (Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\right]$
- $h\left[z^T Q z + w^T R w + (Az + Bw)^T P_t z + z^T P_t (Az + Bw) + z^T \dot{P}_t z\right] = 0$
- Substitute w^*:
  - $\dot{P}_t + A^T P_t + P_t A - P_t B R^{-1} B^T P_t + Q = 0$
  - Or the standard form:
    - $-\dot{P}_t = A^T P_t + P_t A - P_t B R^{-1} B^T P_t + Q$
    - Called Riccati Differential Equation
    - With boundary condition $P_T = Q_f$